TEST REPORT

Lucideon Reference:	N26458 (QT-78966/4/KNA)/Ref. 8
Project Title:	Determination of Water Vapour Transmission Properties - Cup Method in Accordance with BS EN ISO 12572:2016+A1:2014 of L10 Low Carbon Brick
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Report Date:	05 June, 2026
Purchase Order No.:	PO-0003

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1 INTRODUCTION

Earth4Earth Technology Ltd submitted samples of their L10 Low Carbon bricks for Determination of water vapour transmission properties – Cup method in accordance with BS EN ISO 12572:2016+A1:2014.

The testing was performed at the Lucideon Structures Laboratory at Queens Road, Penkhull, Stoke-on-Trent, Staffordshire, ST4 7LQ, UK concluding in May 2026

2 SAMPLES RECEIVED

Solid bricks referenced as L10 Low Carbon were received from the client.

The bricks were nominally 215 mm x 102.5 mm x 65 mm (length x width x height).

The bricks were solid with a tested compressive strength of 50.9 N/mm².

3 TEST METHOD

The determination of water vapour resistance was carried out according to BS EN 12572:2016+A1:2014.

100 mm diameter cores were sampled from the bricks.

The samples were stored at 23°C, 50% relative humidity until a constant mass was achieved.

Conditions were set such that the temperature was at 23°C with the relative humidity being 93% inside the dish (wet state) and the relative humidity being 50% outside the dish and sample (dry state).

An aqueous solution of ammonium dihydrogen phosphate was used to obtain 93% relative humidity.

The test samples were sealed into a glass cup using a wax system with an air gap of 15 mm maintained between the 15 mm of aqueous solution and the test sample, with the sides of the samples protruding above the glass cup using the same wax system.

The samples were stored in an environmental chamber controlled at 23 (± 2)°C, and 50 (± 5)% relative humidity and weighed to 3 No. decimal places at specific time intervals, until 5 No. successive determinations of change in mass per weighing interval for each test sample were obtained.

The mass change rate was calculated and used to calculate the water vapour resistance.

4 RESULTS

Table 1 – Calculation of Difference in Vapour Pressure

	Humidity RH (%)	Temperature (ºC)	Vapour Pressure (Pa)	The Value of Δp
Inside	0.93	23	2611.3	1207
Ambient	0.50	23	1403.9	

Table 2 – Results for the Determination of Earth4Earth Technology L10 Low Carbon Bricks Water Vapour Transmission Properties

Sample No.	Gradient (mg/h)	Area (m^2)	Density of Water Vapour Flow Rate (mg/(m^2h))	Water Vapour Permanence (mg/Pa.m^2·h)	Water Vapour Resistance (Pa.m^2.h/mg)	Water Vapour Permeability (mg/m.h.Pa)
1	3.365	0.0079	428.384	0.355	2.818	0.023
2	3.116	0.0079	396.792	0.329	3.042	0.021
3	5.480	0.0079	697.686	0.578	1.730	0.038
4	6.785	0.0079	863.884	0.716	1.397	0.047
5	6.934	0.0079	882.924	0.732	1.367	0.048
Mean	**5.136**	**0.0079**	**653.93**	**0.542**	**2.071**	**0.035**

To obtain an equivalent air layer thickness (s_d), first the values for δ_p, δ_a and the μ value must be found. Using the mean water vapour permeance (W) and sample thickness (d), δ_p can be calculated with the following equation, given in Part 8.5 of the Standard:

$$\delta_p = W*d$$

Where δ_p was calculated to be 0.009 mg/m*h*Pa.

δ_a can be calculated with the following equation, given in Part 8.6 of the Standard:

$$\delta_a = ((0.086*p_0)/(R_v*T*p))*(T/273)^{1.81}$$

Where p_0 is 1013.25 h.Pa as given in Part 3.2 of the Standard, R_v is 0.000462 N*m/mg.K as given in Part 8.6, T is the temperature of the lab in Kelvin and p is found according to Formula 4 in the Standard. This gives δ_a to be 0.714 mg/m*h*Pa.

The μ value can be found using the following equation, given in Part 8.6 of the Standard:

$$\mu = (\Delta p_v * \delta_a)/(g*d)$$

Where Δp_v is 1207 Pa as given in Table 2 in the Standard. This gives an average μ value of 22.75.

Equivalent air layer thickness (s_d) can now be calculated using the equation, given in Part 8.7 of the Standard:

$$S_d=\mu*d$$

Therefore, for the samples of the Earth4Earth Technology L10 Low Carbon Bricks, the values for each of the calculated properties are as follows:

Table 3 – Calculated Properties

Property	Symbol	Value as Calculated
Density of water vapour flow rate	g	5.136
Water vapour permeance	W_p	0.542mg/(m^2.h.Pa)
Water vapour resistance	Z	2.071 (m^2.h.Pa/mg)
Water vapour permeability	δ_p	0.035 (mg/m.h.Pa)
Water vapour resistance factor	μ	22.75
Water vapour diffusion equivalent air layer thickness	S_d	1.50 (m)

NOTE: The results given in this report apply only to the samples that have been tested.

END OF REPORT